Network Simulation Lab: OSPF Dynamic Routing

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Environment: Cisco Packet Tracer 9.0.1 (Host: Linux Mint 21.3)
Objective: Design, configure, and verify a multi-subnet network using OSPF dynamic routing to achieve full end-to-end connectivity.

Topology Overview

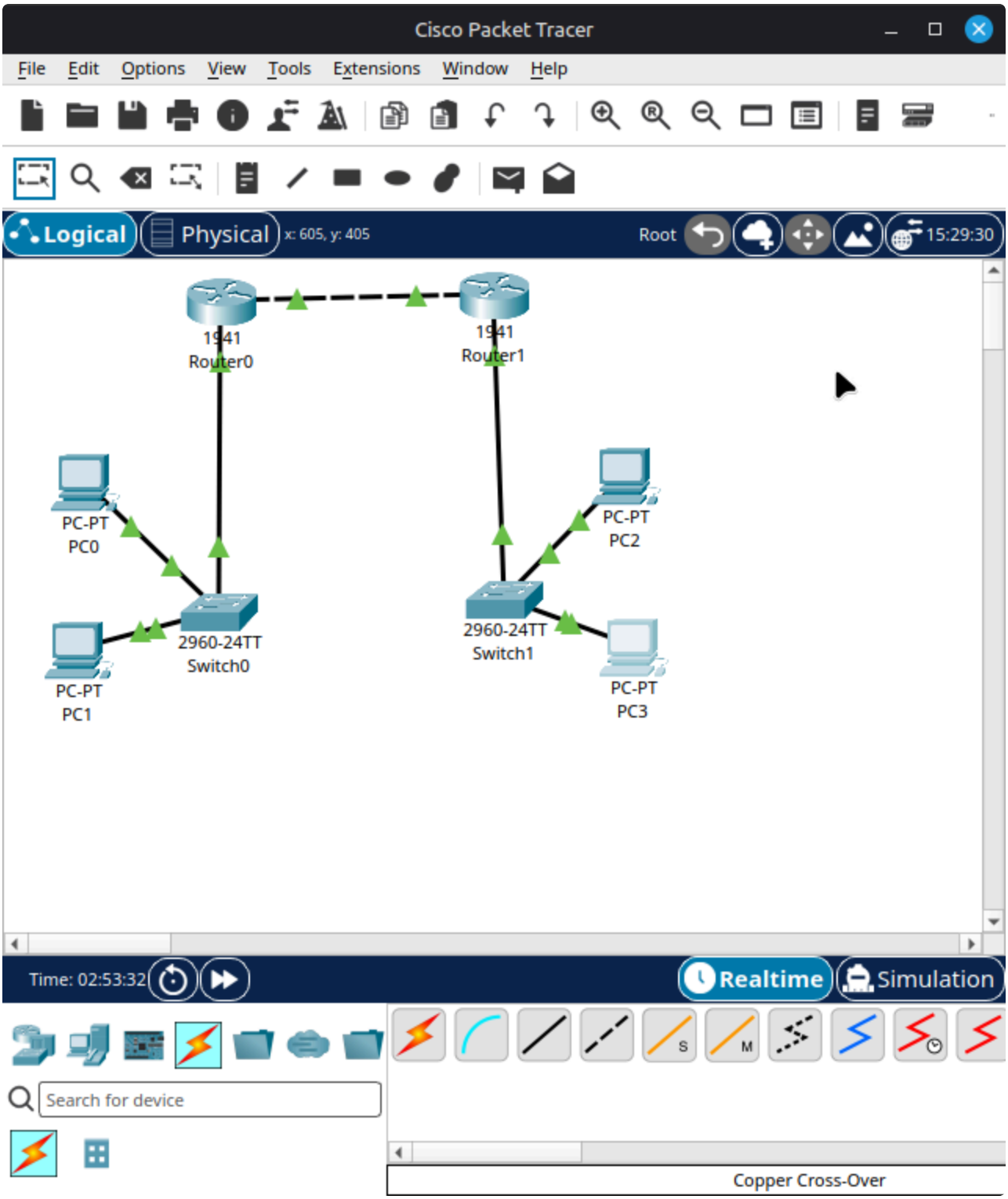
- **2** Cisco 1941 Routers
- **2** Cisco 2960 Switches
- **4** End Devices (PCs)
- **Cabling:** Copper Straight-Through (End devices to Switches, Switches to Routers) and Copper Cross-Over (Router to Router).

IP Addressing Scheme

Subnet	Network Range	Mask	Devices
Left LAN	192.168.1.0	/24 (255.255.255.0)	PC0 (.10), PC1 (.11), Router0 Gi0/0 (.1)
WAN Link	10.0.0.0	/30 (255.255.255.252)	Router0 Gi0/1 (.1), Router1 Gi0/1 (.2)
Right LAN	192.168.2.0	/24 (255.255.255.0)	PC2 (.10), PC3 (.11), Router1 Gi0/0 (.1)

Configuration & Verification

1. Topology & Physical Layer



All devices are connected and powered on. All interfaces have been configured with IP addresses and `no shutdown`, resulting in green link lights across the entire topology.

2. Router Interfaces

```
R0#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
Vlan1                  unassigned      YES NVRAM  administratively down down
R0#
```

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Router0 CLI showing both GigabitEthernet interfaces (`Gi0/0` and `Gi0/1`) with their assigned IP addresses and `up/up` status.

3. OSPF Dynamic Routing

```
R0#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0
O    192.168.2.0/24 [110/2] via 10.0.0.2, 00:58:58, GigabitEthernet0/1

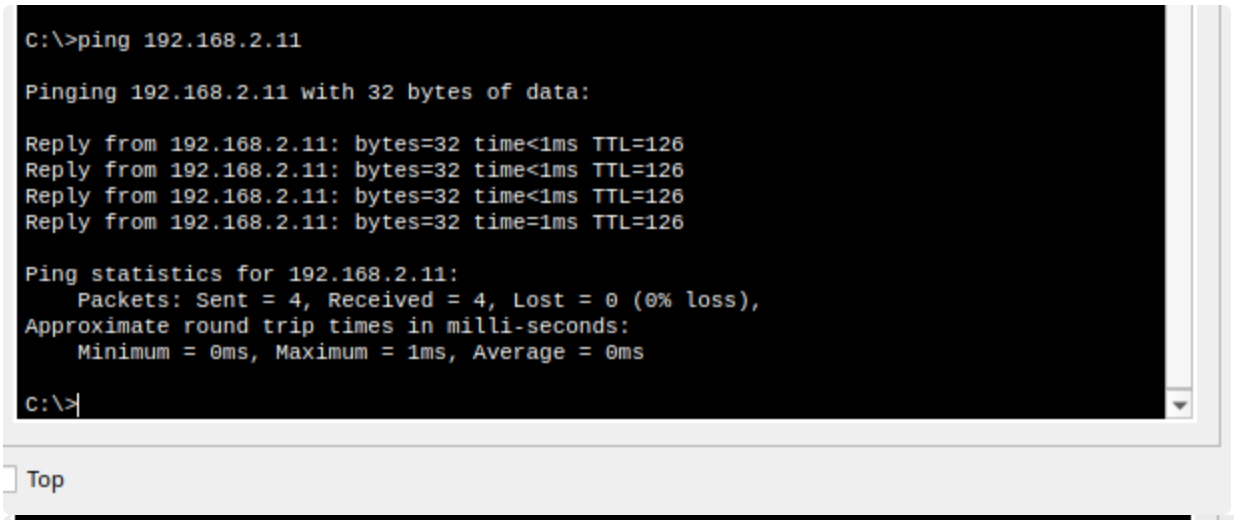
R0#
```

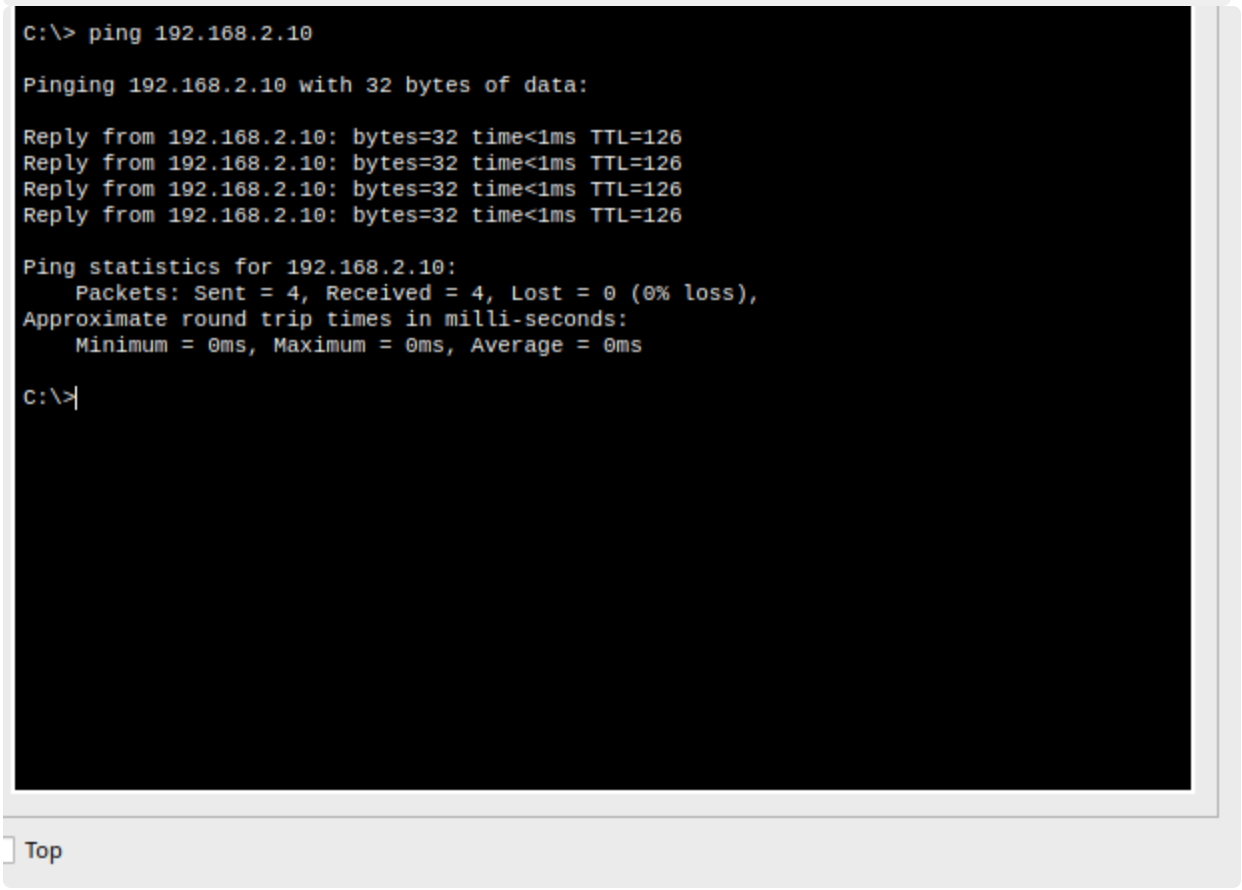
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Router0's routing table showing the right-side LAN (`192.168.2.0/24`) learned dynamically via OSPF (denoted by the `O` code), confirming the routing protocol is exchanging routes successfully.

4. End-to-End Connectivity





Ping tests conducted from PC0 to PC3, and PC1 to PC2, crossing two routers. Successful replies (0% loss, TTL=126) confirm the path is valid and exactly two routers were traversed (TTL starts at 128; 128 - 2 hops = 126).

Key Takeaways

- Successfully implemented a VLSM scheme using /24 for LANs and /30 for the point-to-point WAN link.
- Configured OSPF to avoid manual static routes, ensuring the network can adapt to changes automatically.

- Troubleshot initial interface lockups and CLI configuration issues using tools like `show ip interface brief` and `show ip ospf neighbor`.